

AUTOMATIC DISPENSER Using LM555

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The 555 timer has been a popular choice in electronics for a long time due to its versatility and wide range of applications. It can be used as a timer, oscillator, frequency divider, frequency modulator, triangular signal generator, and more. The 555 timer has three main modes: monostable, astable,

and bistable.

In this article, we will explore a new mode of operation for the 555 timer that generates a unique pulse when the circuit is powered on. The only way to generate another pulse is by turning off the circuit and then powering it on again.

While one might think that this can be achieved with the monostable mode by setting pin 2 to low (ground), doing so would result in the output of the 555 timer always being on. (You can try this yourself.) In this particular monostable mode, pins 5 and 7 are not used. Fig. 1 illustrates the circuit diagram of this

specific monostable circuit.

This new configuration of the 555 timer in monostable mode can be used in various applications, such as alarm systems in cars, turning on a computer in a car, and creating an automatic dispenser for soap or antibacterial gel.

Explanation of operation of this new mode

To explain the operation of this new mode, let's refer to Fig. 1. When the circuit is powered on, capacitor C initially has no voltage, resulting in both pin 6 and pin 2 of the 555 timer having a voltage of zero. In the reset comparator, the inverse input of the comparator has a voltage of $(2/3)V_{CC}$, while the non-inverse input has a voltage of zero. As a result, the internal comparator output is low. Now, let's consider the set comparator. At this time, the inverse input is at 0V, and the non-inverse input is at $(1/3)V_{CC}$. Consequently, the comparator output will be high. In conclusion, when the capacitor has some voltage, the output of the 555 timer will be high.

While the capacitor is charging,

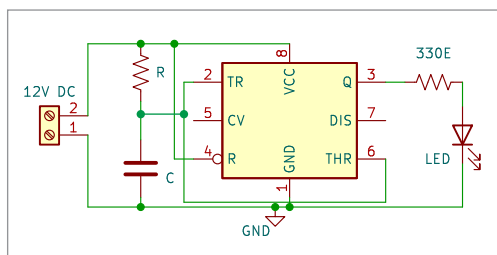


Fig. 1: The specific monostable circuit

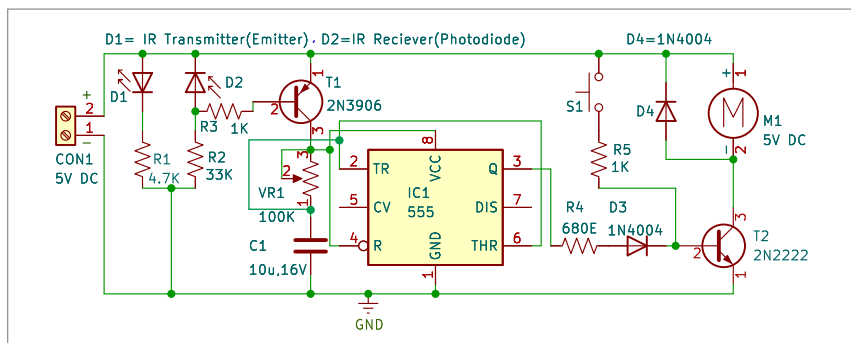


Fig. 2: Circuit diagram of automatic dispenser

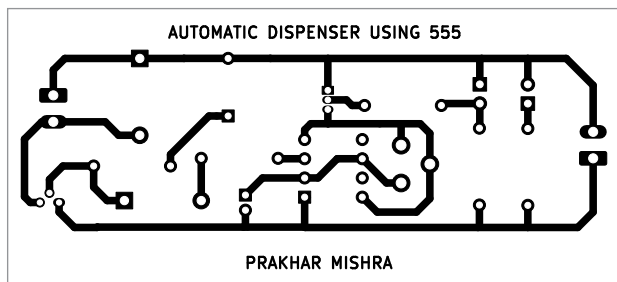


Fig. 3: Actual-size PCB of the circuit

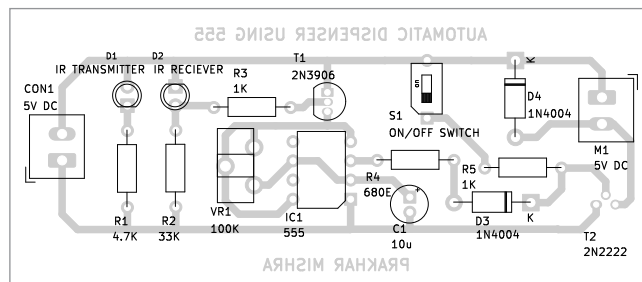


Fig. 4: Component layout of the PCB